THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 153	Intercepts 18 and 17; area $= \frac{1}{2} \times 18 \times 17$.
2	**(c)** 1101	Rate $= \frac{182}{7} = 26$ per km; fixed charge 113.
3	**(b)** 20	Equal slopes: $-\frac{4}{1} = -\frac{k}{5}$, so $k = 20$.
4	**(c)** 17	Equal slopes between consecutive points.
5	**(a)** -4	$8(1) + 2(4) = 16$ gives $k = 8$; slope $= -\frac{k}{2} = -4$.
6	**(a)** 14	x-intercept 5, y-intercept 9.
7	**(a)** 194	$\frac{9}{5} \times 90 + 32$.
8	**(b)** $-\frac{1}{36}$	$\frac{23 - (22)}{-24 - (12)} = -\frac{1}{36}$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$3(2) + 2(3) = 12$, so the pair satisfies the equation; R is the definition used.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; R describes the shape of the graph but does not by itself say why there are infinitely many solutions.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substitute: $k(1) + 3(6) = 27$ gives $k = 9$; Line: $9x + 3y = 27$; $y = -3x + \frac{27}{3}$ (1); Slope $= -\frac{9}{3} = -3$; y-intercept $= \frac{27}{3}$ (1).	1+1
12	x-intercept: put $y = 0$: $9x = 135$, so $x = 15$; y-intercept: put $x = 0$: $15y = 135$, so $y = 9$ (1); Area $= \frac{1}{2} \times 15 \times 9 = \frac{135}{2}$ square units (1).	1+1
13	Slope $= \frac{-21 - (-24)}{-9 - (-12)} = 1$; Use $(-12, -24)$: $-24 = 1(-12) + c$ (1); $c = -12$; $y = x - 12$ (1).	1+1
14	Substitute: $k(3) + 4(6) = 36$ gives $k = 4$; Line: $4x + 4y = 36$; $y = -x + \frac{36}{4}$ (1); Slope $= -\frac{4}{4} = -1$; y-intercept $= \frac{36}{4}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Slope $= \frac{17 - 0}{0 - 18} = -\frac{17}{18}$; Equation: $y = -\frac{17}{18}x + 17$, i.e. $17x + 18y = 306$ (1); Check: (18, 0)$gives$17\times18=306 (1); $Area$=\frac12\times18\times17=153$ square units (1).	1+1+1
16	Rate $= \frac{105}{7} = 15$ per kL; Fixed charge $= 47$; equation $y = 15x + 47$ (1); $317 = 15x + 47$ gives $x = 18$ kL (1); Bill for 0 kL $= 47$, the y-intercept (1).	1+1+1
17	Slope of the first two points $= \frac{-13 - (-9)}{21 - (19)} = -2$ (1); Slope between $(21, -13)$ and $(23, k)$ must also be -2: $\frac{k - (-13)}{2} = -2$ (1); $k = -13 - 2 \times 2 = -17$ (1).	1+1+1
	OR: Slope $= \frac{18 - 0}{0 - 15} = -\frac{18}{15}$; Equation: $y = -\frac{18}{15}x + 18$, i.e. $18x + 15y = 270$ (1); Check: (15, 0)$gives$18\times15=270 (1); $Area$=\frac12\times15\times18=135$ square units (1).	1+1+1
18	Rate $= \frac{96}{6} = 16$ per kL; Fixed charge $= 52$; equation $y = 16x + 52$ (1); $324 = 16x + 52$ gives $x = 17$ kL (1); Bill for 0 kL $= 52$, the y-intercept (1).	1+1+1
	OR: Slope of the first two points $= \frac{-15 - (-17)}{-17 - (-18)} = 2$ (1); Slope between $(-17, -15)$ and $(-16, k)$ must also be 2: $\frac{k - (-15)}{1} = 2$ (1); $k = -15 + 2 \times 1 = -13$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	(i) $5x + 20y = 100$ (1). (ii) $5(4) + 20(4) = 100$, so yes (1). (iii) $5x + 40 = 100 \Rightarrow x = 12$ pens (1). (iv) x-intercept $= 20$ (put $y = 0$): 20 pens and no notebook (1); y-intercept $= 5$ (put $x = 0$): 5 notebooks and no pen (1).	1+1+1+2
20	Slope $= \dfrac{6-(2)}{1-(0)} = 4$; $y = 4x + 2$ (1); $30 = 4k + 2$ gives $k = 7$ (2); x-intercept: $y = 0$ gives $x = -\frac{1}{2}$ (2).	1+2+2
	OR: Rate $= \dfrac{310 - 222}{8 - 4} = 22$ per km (1); Fixed charge $= 222 - 22 \times 4 = 134$ (1); Fare for 38 km $= 22 \times 38 + 134$ (1); $=$ ₹970 (2).	1+1+1+2

Mistakes to avoid:

- Finding the x-intercept by putting $x = 0$ (that is the y-intercept); for the x-intercept put $y = 0$.
- Writing the slope of $ax + by = c$ as $\frac{a}{b}$ instead of $-\frac{a}{b}$.
- Saying a linear equation in two variables has one solution; it has infinitely many.
- Treating $x = 5$ as a point; in the plane it is a vertical line.
- Plotting the pair as (y, x) instead of (x, y).
- Making a sign slip while transposing, e.g. $2x + y = 8 \Rightarrow y = 2x + 8$ instead of $y = -2x + 8$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 20	Equal slopes: $-\frac{4}{3} = -\frac{k}{15}$, so $k = 20$.
2	**(b)** 21	Equal slopes between consecutive points.
3	**(d)** -5	$10(1) + 2(1) = 12$ gives $k = 10$; slope $= -\frac{k}{2} = -5$.
4	**(a)** 25	x-intercept 18, y-intercept 7.
5	**(a)** 185	$\frac{9}{5} \times 85 + 32$.
6	**(a)** $-\frac{7}{18}$	$\frac{-5-(-12)}{4-(22)} = -\frac{7}{18}$.
7	**(c)** 253	Intercepts 23 and 22; area $= \frac{1}{2} \times 23 \times 22$.
8	**(d)** 2055	Rate $= \frac{288}{6} = 48$ per km; fixed charge 135.
9	**(c)** A is true but R is false.	Putting $y = 0$ gives $x = 6$, so A is true; R is false because points on the x-axis have y-coordinate 0.
10	**(d)** A is false but R is true.	A is false because $x = 5$ gives the line through $(5, 0), (5, 1), (5, -2), \ldots$; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	x-intercept: put $y = 0$: $15x = 285$, so $x = 19$; y-intercept: put $x = 0$: $19y = 285$, so $y = 15$ (1); Area $= \frac{1}{2} \times 19 \times 15 = \frac{285}{2}$ square units (1).	1+1
12	Slope $= \frac{-19-(5)}{-9-(3)} = 2$; Use $(3, 5)$: $5 = 2(3) + c$ (1); $c = -1$; $y = 2x - 1$ (1).	1+1
13	Substitute: $k(2) + 5(-2) = 10$ gives $k = 10$; Line: $10x + 5y = 10$; $y = -2x + \frac{10}{5}$ (1); Slope $= -\frac{10}{5} = -2$; y-intercept $= \frac{10}{5}$ (1).	1+1
14	x-intercept: put $y = 0$: $16x = 240$, so $x = 15$; y-intercept: put $x = 0$: $15y = 240$, so $y = 16$ (1); Area $= \frac{1}{2} \times 15 \times 16 = 120$ square units (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Slope $= \frac{7-0}{0-12} = -\frac{7}{12}$; Equation: $y = -\frac{7}{12}x + 7$, i.e. $7x + 12y = 84$ (1); Check: (12, 0)$gives$7\times12=84(1); $Area$ =\frac12\times12\times7=42$ square units (1).	1+1+1
16	Rate $= \frac{88}{11} = 8$ per kL; Fixed charge $= 30$; equation $y = 8x + 30$ (1); $206 = 8x + 30$ gives $x = 22$ kL (1); Bill for 0 kL $= 30$, the y-intercept (1).	1+1+1
17	Slope of the first two points $= \frac{32-(30)}{-25-(-26)} = 2$ (1); Slope between $(-25, 32)$ and $(-24, k)$ must also be 2: $\frac{k-(32)}{1} = 2$ (1); $k = 32 + 2 \times 1 = 34$ (1).	1+1+1
	OR: Slope $= \frac{13-0}{0-6} = -\frac{13}{6}$; Equation: $y = -\frac{13}{6}x + 13$, i.e. $13x + 6y = 78$ (1); Check: (6, 0)$gives$13\times6=78 (1); $Area$=\frac12\times6\times13=39$ square units (1).	1+1+1
18	Rate $= \frac{130}{10} = 13$ per kL; Fixed charge $= 38$; equation $y = 13x + 38$ (1); $285 = 13x + 38$ gives $x = 19$ kL (1); Bill for 0 kL $= 38$, the y-intercept (1).	1+1+1
	OR: Slope of the first two points $= \frac{-21-(-19)}{0-(-1)} = -2$ (1); Slope between $(0, -21)$ and $(1, k)$ must also be -2: $\frac{k-(-21)}{1} = -2$ (1); $k = -21 - 2 \times 1 = -23$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	(i) $y = -2x + 8$; slope $= -2$ (1). (ii) 8, i.e. the point $(0, 8)$ (1). (iii) $2x = 8 \Rightarrow x = 4$, i.e. $(4, 0)$ (1). (iv) $2(3) + 2 = 8$, so $(3, 2)$ lies on it (1); at $x = -1$: $y = 8 + 2 = 10$ (1).	1+1+1+2
20	Slope $= \dfrac{8 - (0)}{1 - (-3)} = 2$; $y = 2x + 6$ (1); $20 = 2k + 6$ gives $k = 7$ (2); x-intercept: $y = 0$ gives $x = -3$ (2).	1+2+2
	OR: Rate $= \dfrac{618 - 178}{12 - 2} = 44$ per km (1); Fixed charge $= 178 - 44 \times 2 = 90$ (1); Fare for 20 km $= 44 \times 20 + 90$ (1); $=$ ₹970 (2).	1+1+1+2

Mistakes to avoid:

- Finding the x-intercept by putting $x = 0$ (that is the y-intercept); for the x-intercept put $y = 0$.
- Writing the slope of $ax + by = c$ as $\frac{a}{b}$ instead of $-\frac{a}{b}$.
- Saying a linear equation in two variables has one solution; it has infinitely many.
- Treating $x = 5$ as a point; in the plane it is a vertical line.
- Plotting the pair as (y, x) instead of (x, y).
- Making a sign slip while transposing, e.g. $2x + y = 8 \Rightarrow y = 2x + 8$ instead of $y = -2x + 8$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** -4	$12(1)+3(-4)=0$ gives $k=12$; slope $=-\frac{k}{3}=-4$.
2	**(b)** 23	x-intercept 7, y-intercept 16.
3	**(d)** 239	$\frac{9}{5}\times 115+32$.
4	**(d)** $-\frac{5}{12}$	$\frac{-1-(9)}{17-(-7)}=-\frac{5}{12}$.
5	**(d)** 187	Intercepts 22 and 17; area $=\frac{1}{2}\times 22\times 17$.
6	**(d)** 1319	Rate $=\frac{216}{6}=36$ per km; fixed charge 59.
7	**(c)** 5	Equal slopes: $-\frac{1}{7}=-\frac{k}{35}$, so $k=5$.
8	**(b)** 5	Equal slopes between consecutive points.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$y=-3x+7$ has slope -3, found as R describes.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	At $x=0$, $y=0$, so A is true; R is true, but the slope does not decide that the line passes through the origin.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Slope $=\frac{11-(-6)}{4-(-13)}=1$; Use $(-13,-6)$: $-6=1(-13)+c$ (1); $c=7$; $y=x+7$ (1).	1+1
12	Substitute: $k(1)+3(1)=15$ gives $k=12$; Line: $12x+3y=15$; $y=-4x+\frac{15}{3}$ (1); Slope $=-\frac{12}{3}=-4$; y-intercept $=\frac{15}{3}$ (1).	1+1
13	x-intercept: put $y=0$: $21x=504$, so $x=24$; y-intercept: put $x=0$: $24y=504$, so $y=21$ (1); Area $=\frac{1}{2}\times 24\times 21=252$ square units (1).	1+1
14	Slope $=\frac{10-(-25)}{-1-(-8)}=5$; Use $(-8,-25)$: $-25=5(-8)+c$ (1); $c=15$; $y=5x+15$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Slope $=\frac{7-0}{0-9}=-\frac{7}{9}$; Equation: $y=-\frac{7}{9}x+7$, i.e. $7x+9y=63$ (1); Check: (9, 0)$gives$7\times9=63(1); $Area$ =\frac12\times9\times7=\frac{63}{2}$ square units (1).	1+1+1
16	Rate $=\frac{48}{6}=8$ per kL; Fixed charge $=40$; equation $y=8x+40$ (1); $192=8x+40$ gives $x=19$ kL (1); Bill for 0 kL $=40$, the y-intercept (1).	1+1+1
17	Slope of the first two points $=\frac{9-(15)}{21-(19)}=-3$ (1); Slope between $(21,9)$ and $(23,k)$ must also be -3: $\frac{k-(9)}{2}=-3$ (1); $k=9-3\times 2=3$ (1).	1+1+1
	OR: Slope $=\frac{13-0}{0-15}=-\frac{13}{15}$; Equation: $y=-\frac{13}{15}x+13$, i.e. $13x+15y=195$ (1); Check: (15, 0)$gives$13\times15=195 (1); $Area$=\frac12\times15\times13=\frac{195}{2}$ square units (1).	1+1+1
18	Rate $=\frac{56}{7}=8$ per kL; Fixed charge $=85$; equation $y=8x+85$ (1); $237=8x+85$ gives $x=19$ kL (1); Bill for 0 kL $=85$, the y-intercept (1).	1+1+1
	OR: Slope of the first two points $=\frac{-30-(-26)}{-18-(-20)}=-2$ (1); Slope between $(-18,-30)$ and $(-16,k)$ must also be -2: $\frac{k-(-30)}{2}=-2$ (1); $k=-30-2\times 2=-34$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	(i) $2l + 2b = 30$, i.e. $l + b = 15$ (1). (ii) $b = 6$ m (1). (iii) $10 + 6 = 16 \neq 15$, so no (1). (iv) $l = 1, 2, \ldots, 14$ gives 14 solutions (1); intercepts $(15, 0)$ and $(0, 15)$ (1).	1+1+1+2
20	Slope $= \dfrac{4-(-8)}{8-(-4)} = 1$; $y = x - 4$ (1); $3 = 1k - 4$ gives $k = 7$ (2); x-intercept: $y = 0$ gives $x = 4$ (2).	1+2+2
	OR: Rate $= \dfrac{463 - 313}{8 - 5} = 50$ per km (1); Fixed charge $= 313 - 50 \times 5 = 63$ (1); Fare for 37 km $= 50 \times 37 + 63$ (1); $=$ ₹1913 (2).	1+1+1+2

Mistakes to avoid:

- Finding the x-intercept by putting $x = 0$ (that is the y-intercept); for the x-intercept put $y = 0$.
- Writing the slope of $ax + by = c$ as $\frac{a}{b}$ instead of $-\frac{a}{b}$.
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- Plotting the pair as (y, x) instead of (x, y).
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